



Education

Advancing project leadership education through AI-enhanced game-based learning

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ABSTRACT

This paper offers a design-led investigation of AI-enhanced game-based learning (AI-GBL) as a transformative pedagogical approach for project leadership education. Addressing the persistent theory–practice gap in project management training, the study incorporates generative AI into ethically complex, emotionally engaging simulation environments that mimic real-world stakeholder challenges. The intervention allows learners to participate in adaptive decision-making, critical reflection, and ethical reasoning in face-to-face and blended formats, based on principles from experiential learning, productive failure, and relational pedagogy.

Implemented in 2024 across undergraduate and postgraduate cohorts at an Irish university, the intervention was examined through six interconnected design themes: relevance, agility, identity, ethics, motivation, and adaptability. These themes were instructional principles and analytical lenses, revealing significant conceptual learning gains, enhanced ethical deliberation, and evidence of evolving leadership identity. Qualitative data from reflective journals and digital dialogues demonstrated students' growing capacity for situated judgement, collaborative reasoning, and moral awareness.

This paper offers a conceptual roadmap for integrating AI-GBL into leadership education, viewing generative AI as a pedagogical co-ordinator that supports ambiguity, highlights values, and fosters human-centred learning. It advocates for ethically designed AI to develop project leaders who are skilled, self-aware, ethically grounded, and capable of leading with integrity in an era of complex technology. As AI reshapes work and learning, AI-GBL provides a way to align human abilities with technological potential.

1. Introduction

Preparing competent project-management professionals remains a pressing concern for both academia and industry. The gap between theoretical instruction and the practical demands of complex, adaptive, and ethically charged project environments is a persistent challenge. While students often learn methodologies and tools, they struggle to transfer this knowledge into real-world contexts marked by ambiguity and fluid stakeholder dynamics. This disconnect is recognised widely as the “theory–practice gap” (Darrell et al., 2010; Egginton, 2012).

The growing demand for project managers and the persistent phenomenon of the accidental project manager, typically individuals placed in roles based on general managerial or technical expertise rather than formal project-management education, has further widened the gap. Conventional project-management education often prioritises routine expertise, focusing on the efficient application of established tools and

methods, while providing limited support for developing adaptive expertise, which requires innovation, contextual judgement, and the capacity to operate effectively under conditions of uncertainty (Pelgrim et al., 2022; Ward et al., 2018). As Huemann and Turner observe, “Complex projects are mysteries which require innovation. Conventional project management is not good at innovation” (Turner and Huemann, 2024).

Addressing these conditions requires educational approaches that promote reflection, decision-making under pressure, and ethical reasoning, the capacities that traditional teaching methods often fail to support at scale. For the first time, generative AI offers a breakthrough in designing and delivering such learning experiences. Through customisable, responsive, and emotionally resonant scenario-based environments, AI-GBL enables students and practitioners to rehearse complex project challenges, navigate moral ambiguity, and refine their leadership identity within psychologically safe yet realistic simulations.

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This study proposes that AI-GBL is not merely a pedagogical enhancement but a paradigm shift. It positions intelligent systems not as content providers, but as reflective partners capable of scaffolding adaptive expertise, accelerating situated learning, and embedding ethical deliberation into the learning process. What was once aspirational, real-time, high-stakes leadership rehearsal is now feasible, scalable, and customisable across contexts. This study explores how such environments can help cultivate project leaders who are not only technically skilled but also narratively self-aware, ethically grounded, and prepared to lead in an era of intelligent complexity.

This paper presents an experimental intervention built around the AI-GBL framework. Game-based learning (GBL) provides interactive environments where learners engage with realistic scenarios and experience consequences in low-risk settings (Plass et al., 2015; Zakeri et al., 2023). Serious games extend beyond motivation, fostering critical thinking, resilience, and communication skills central to project leadership. In this context, GBL can simulate real-world projects' emotionally charged and time-sensitive nature, offering "stress inoculation training."

The model enhances GBL through generative AI (GenAI), enabling real-time scenarios, personalised feedback, and dynamic narrative branching adaptation. It draws on Kolb's experiential learning theory and is structured using the 5E instructional model (Engage, Explore, Explain, Elaborate, Evaluate), providing a scaffolded, socially embedded learning cycle (Kolb & Kolb, 2005, 2008, 2017; Owens and Sadler, 2024). The design also incorporates the concept of productive failure, where students encounter disorienting challenges that stimulate reflection and deeper learning (Kapur, 2016).

GenAI serves both as a content generator and a pedagogical agent. It can introduce ethical dilemmas, interpret learner behaviour, and adapt real-life scenarios to match learners' needs. This approach aligns with current research on AI-driven scaffolding and personalised learning (Wang et al., 2023; Zawacki-Richter et al., 2019), particularly in fostering ethical development and promoting inclusive, learner-centred educational experiences.

The intervention also uses virtual learning communities to support identity development and ethical deliberation. These communities enable peer feedback, collaborative inquiry, and dialogic learning as key components of relational pedagogy (Chadha, 2024).

Six core themes underpin the AI-GBL model: relevance, agility, identity, ethics, motivation, and adaptability. These themes are design principles and analytical lenses for examining how AI can transform project leadership education. Rather than presenting a fixed framework, we offer these themes as adaptable tools to guide educational transformation in the context of rapid AI advancement (Kamalov et al., 2023).

Adoption of AI in education continues to face barriers, not due to technical limitations, but because of trust deficits, skill shortages, and weak alignment with pedagogical theory (Jöhnk et al., 2020; Zawacki-Richter et al., 2019). Our approach treats AI not as an efficiency tool but as a reflective ally, aligned with calls for human-centred, ethically grounded design (Shneiderman, 2020).

This study explores how AI-GBL can support the formation of project leaders who are not only technically skilled but also adaptive, ethically literate, and prepared to lead in uncertain and dynamic environments. While much of the literature on AI in education focuses on automation or personalised content delivery, this study proposes a broader pedagogical role for AI in shaping ethical, emotional, and identity development in future project leaders. Table 1 outlines the study's four main conceptual contributions, which frame AI-GBL as a transformative, not merely technological, innovation.

This study positions the six pedagogical themes as actionable design anchors, not abstract ideals. Each theme targets a specific leadership capability that AI-GBL environments actively cultivate. These capabilities are crucial when automation increasingly handles routine tasks, and human leadership is defined by judgement, reflection, and

Table 1
Conceptual contributions of the study.

Contribution	Description	Why It Matters
AI as Pedagogical Co-Orchestrator	Reframes generative AI as a dynamic, responsive agent that shapes learning through feedback, scenario adaptation, and reflection prompts.	Moves beyond efficiency narratives to human-centred, real-time learning facilitation.
Virtue Ethics and Phronesis in Design	Integrates ethical reasoning as a core design principle grounded in practical wisdom, rather than rule-based decision-making.	Elevates ethical-AI literacy as a leadership capability, not just a compliance issue.
Narrative Identity Formation	Uses immersive decision-making and reflection to help learners construct and revise their leadership identity.	Aligns learning with real-world leadership development processes and values-based action.
Failure and Ambiguity as Learning Catalysts	Embeds productive failure, emotional tension, and scenario ambiguity to stimulate deeper learning.	Reorients education away from control and correctness toward adaptive judgement.

relational intelligence. Table 2 outlines how each theme maps to distinct learning capacities essential for resilient, ethical, and future-ready project leadership.

The research focus explores how AI-GBL can advance project leadership education in ethically complex, real-world contexts. Three questions guide it:

1. How does AI-GBL support the development of adaptive, ethical, and reflexive leadership capabilities?
2. How do the six core pedagogical themes inform learner experience and design?
3. What observable learning outcomes emerge from student engagement with AI-mediated simulations?

These questions shape the study's conceptual and empirical contributions, offering a roadmap for leveraging AI to foster leadership capacities beyond procedural competence.

2. Research methodology

This study employed a design-based, mixed-methods approach, integrated into two project management modules in 2024 at an Irish university. The intervention was embedded in regular teaching and assessment cycles, requiring no additional staffing or resources. Two cohorts participated: 113 undergraduates engaged in weekly face-to-face tutorials featuring AI-generated project scenarios, and forty-seven postgraduates who participated in blended learning, combining asynchronous scenario-based challenges and peer feedback activities via the Brightspace virtual learning environment (VLE) with scheduled in-class tutorials focused on collaborative scenario work.

We collected quantitative data for undergraduate students through pre- and post-assessments of conceptual understanding. We used descriptive statistics to summarise learning gains and applied a t-test to compare outcomes between high- and low-engagement groups. We defined undergraduate engagement based on participation in weekly AI scenario challenges, contributions to in-class discussions, and tutor-recorded attendance and task completion, although we did not analyse formal VLE logs quantitatively; tutor observations and submitted coursework served as practical proxies for assessing engagement levels.

We collected qualitative data from weekly tutor reflections, anonymised discussion board contributions, and reflective journals. The teaching team led a collaborative thematic analysis of these sources. We initially aligned codes deductively with the six design themes and refined them iteratively using inductive insights. To ensure coding

Table 2
Themes and associated learning outcomes in project leadership education.

Theme	Learning Outcome	Why It is Relevant
1. Relevance	Apply tools and frameworks in ethically and socially complex project settings.	Responds to the theory–practice gap and rising need for phronetic, contextualised leadership.
2. Agility	Make timely decisions amid ambiguity and rapidly changing conditions.	Reflects project volatility and the shift from control to adaptation in leadership models.
3. Identity	Construct and reflect on one’s leadership values and positionality.	Supports leadership as a relational and narrative process, not just task execution.
4. Ethics	Critically evaluate AI advice and make values-based, accountable decisions.	Urgent in AI-integrated work; supports ethical AI literacy and moral agency in uncertain contexts.
5. Motivation	Sustain intrinsic engagement and forward momentum through challenge and reflection.	Necessary for resilience, long-term growth, and learner agency in complex learning environments.
6. Adaptability	Lead through failure, adjust strategies, and learn iteratively.	Enables adaptive expertise as essential for leading novel, dynamic, and unpredictable projects.

consistency, we conducted peer cross-checking. We supported triangulation by aligning findings across tutor notes, learner reflections, and discussion threads.

Throughout this study, we adopted a reflexive orientation, recognising our dual roles as academic educators and experienced project professionals. This positionality shaped the intervention’s design and delivery, informed by our direct engagement with leadership development, motivational learning, and complex project environments. Drawing on this perspective, we translated theoretical constructs such as ethical reasoning, narrative identity, and productive failure into pedagogically grounded, professionally relevant experiences. We treated the classroom not as a neutral site of content delivery, but as a space for identity rehearsal, ethical deliberation, and decision-making under ambiguity. We introduced the AI system not as an instructional novelty, but as a co-facilitator of moral reflection and adaptive challenge. At each stage, our design decisions reflected a commitment to learner agency, contextual integrity, and the development of leadership capabilities that extend beyond procedural competence.

We fully complied with the university’s ethics policy for pedagogical research. We did not require formal approval, as we collected all data from standard coursework activities, anonymised the information, and included it in the study only with participants’ voluntary, opt-in consent. Our approach ensured that no student assessment outcomes were affected.

3. Thematic Framework and theoretical groundings

Emerging through an iterative, design-based process, these themes serve as guiding principles and analytic lenses for the AI-GBL intervention. Rather than remaining abstract, each theme is translated into specific learning outcomes (see Table 2), offering a practical foundation for designing AI-enhanced environments that foster adaptive expertise, narrative identity, and ethical judgement in future project leaders.

This framework draws on experiential learning theory (Kolb and Kolb, 2008), productive failure (Kapur, 2013), narrative identity (McAdams and McLean, 2013), and virtue ethics (Eikeland, 2012) to embed reflection, complexity, and moral reasoning into every layer of the learning design. Together, the six themes form a foundation for a pedagogical approach that views project leadership not merely as a skillset, but as a dynamic, ethical practice of becoming.

3.1. Theme 1 - relevance

Despite the rising trend of Projectification, project management education remains profoundly disconnected from practical demands. Tekic et al. (2022) emphasises this, revealing that practitioners co-author fewer than 2 % of project management research articles as evidence of an epistemic silo that isolates academia from its applied roots. Dalcher (2016) also critiques this disconnection, arguing that education continues to produce “trained technicians”: graduates skilled in executing tools and templates but unprepared for the complexity, uncertainty, and stakeholder dynamics that characterise real projects.

This failure of *relevance* is structural. Bredillet (2014) attributes it to prioritising abstract rigour over contextual wisdom. Dominant discourses in project management gravitate towards control and prediction, favouring instrumental rationality over reflective practice. As Turner and Huemann (2024) observe, “Complex projects are mysteries which require innovation. Conventional project management is not good at innovation.”

This study uses praxis, defined by Bredillet (2010) as “knowing-in-action”: a fusion of reflection and action embedded within authentic contexts. This Aristotelian perspective, grounded in phronēsis (practical wisdom), frames ethical decision-making as situated, intuitive, and emotionally intelligent. Eikeland (2012) asserts that praxis cannot be reduced to technocratic or relativist models; it requires responsive, human-centred judgment.

In this context, we introduced AI-GBL as a pedagogical engine for praxis, designed not to replace human judgement but to surface complexity, ambiguity, and ethical tension. The large language model (LLM) scaffolded the learning process by prompting reflection-in-action and generating branching dilemmas that simulated crisis-inflicted project environments, contexts where checklist logic falters and stakeholder alignment becomes contested. Educators retained design authority and ethical oversight throughout, using the 5E instructional model (Engage, Explore, Explain, Elaborate, Evaluate) to align scenario-based learning with experiential pedagogical principles (Hopkinson and Hogg, 2004).

A quasi-experimental study conducted in 2024 evaluated this intervention across 160 undergraduate and postgraduate students. While all participants completed a baseline test, only the experimental group engaged in AI-GBL sessions embedded in their VLEs. These learners outperformed the control group (traditional case-based learning) by an average of sixteen percentage points in post-test scores ($t(111) = 9.13$, $p < .001$). More revealing were qualitative shifts: postgraduate reflective journals included richer ethical vocabulary and explicit references to virtue ethics and stakeholder trade-offs. Linguistic-inquiry analyses also detected increased use of first-person plural pronouns (“we decided”, “we negotiated”), indicating emergent collective identity and alignment with identity-leadership principles (Haslam et al., 2022). However, some students admitted to an “over-reliance on the AI’s first suggestion” when tutor scaffolding was absent.

These findings support the thesis that AI-mediated, narrative-rich rehearsal can catalyse praxis. AI-GBL bridges the theory–practice divide by confronting students with ambiguity, ethical dilemmas, and contested stakeholder perspectives in a safe yet realistic environment. It also aligns with flow theory, which posits that optimal learning occurs at the edge of complexity (Csikszentmihalyi and Csikszentmihalyi, 1990). Where challenge meets engagement, competencies emerge that underpin adaptive expertise and resilient leadership.

While these findings are encouraging, they must be interpreted within the practical boundaries of a live teaching intervention. Although practitioner input informed scenario design, their limited involvement as co-researchers means this study only partially fulfils Tekic’s (2022) call for more deeply engaged scholarship. Additionally, several learners demonstrated an over-reliance on AI-generated suggestions without tutor scaffolding, highlighting the need to advance ethical AI literacy as a core component of professional competence. These observations reinforce a central argument: relevance in AI-GBL environments is not an incidental outcome but a necessary design imperative, requiring co-creation with practitioners and intentional alignment with contemporary project leadership’s ethical and adaptive demands.

3.2. Theme 2 - agility

While Theme 1 addressed how ethical praxis restores relevance in project leadership education, Theme 2 turns to *agility*, not as a fixed learner trait but as a capability actively developed through structured challenge, reflection, and feedback. This theme aligns with contemporary understandings of project environments as volatile, uncertain, complex, and ambiguous (VUCA), where leadership depends on adaptive responsiveness rather than static expertise (Hillson, 2024).

The AI-GBL model scaffolded this development through scenario-based sessions that disrupted procedural habits. Designed stressors, including regulatory reversals, stakeholder conflict, or budget volatility, created structured difficulty and productive struggle. The goal was to simulate learning at the edge of competence, where ambiguity and stretch trigger deeper engagement and cognitive restructuring (Csikszentmihalyi and Csikszentmihalyi, 1990; Kapur, 2016).

Where Theme 1 highlighted ethical reasoning as an emergent capability, the same data revealed a parallel shift toward adaptive reasoning. Reflective journals showed how learners transitioned from linear problem-solving to iterative framing. Several students described their

actions as “course correcting” or “re-evaluating based on new stakeholder input”, illustrating increasing cognitive flexibility. These findings reinforce the link between scaffolded complexity and the cultivation of agile leadership, as articulated in adaptive expertise research (Green et al., 2015; Paas and Merriënboer, 1994; Rangavittal, 2023).

Although this study did not apply formal instruments to measure cognitive load, such as the NASA Task Load Index (NASA TLX) (Hart and Staveland, 1988), prior research highlights its value as a multidimensional tool for assessing cognitive effort, frustration, and perceived workload. In this implementation, tutor observations, including signs of hesitation, clarification requests, or visible fatigue, served as practical and context-sensitive proxies for monitoring learner stretch. While less precise than validated tools, this approach reflects growing interest in learner-sensitive instrumentation that adapts to real-time behavioural cues (Kamalov et al., 2023; Zawacki-Richter et al., 2019). These insights point to future opportunities to incorporate multimodal measures such as gaze tracking or biometric feedback into AI-enhanced environments to support more finely tuned challenge calibration.

As AI-enhanced platforms evolve, this principle invites the integration of multimodal indicators like gaze tracking, biometric feedback, or affective computing to calibrate the challenge dynamically. Such responsiveness supports what Kamalov et al. (2023) describe as co-orchestrated learning, in which intelligent systems and educators collaboratively maintain the learner’s zone of productive difficulty. An example is detecting a user showing signs of cognitive overload through webcam-based facial recognition, reducing the next challenge’s complexity, or prompting a reflective pause.

Theme 2 ultimately asserts that agility is not a passive outcome of exposure to complexity; it is a capability that must be designed for, rehearsed, and reflected upon. When students encounter structured ambiguity in psychologically safe environments, they learn to manage disruption and lead through it.

3.3. Theme 3 – identity formation

The previous themes addressed the recovery of relevance through praxis and the cultivation of agility through structured struggle. This theme shifts focus to leadership identity as a central educational outcome. Effective project leadership is not solely defined by procedural fluency or domain knowledge but by a reflexive understanding of oneself as a leader, particularly amid ethical complexity and social negotiation.

At the heart of this approach lies the construct of narrative identity, defined as an internalised and evolving life story that weaves past experiences with imagined futures to create meaning and continuity (McAdams and McLean, 2013). In this view, leadership development becomes a form of meaning-making in which individuals author their identities by reflecting on morally significant moments. Values are not abstract concepts but lived commitments emerging from situated judgement and narrative coherence.

This perspective aligns with contemporary project environments, where leadership is increasingly viewed not as a fixed role, but as a dynamic, negotiated identity shaped through interaction and shared influence (Müller, 2024). Leadership responsibilities may shift between vertical, horizontal, and distributed forms, inviting project professionals to continuously reframe their role and interpret complex social signals within transient project structures.

The Project Leadership Model reinforces this identity-centric view by foregrounding sensemaking, projecting, and judging as core leadership acts, often articulated through narrative reasoning (Sergeeva et al., 2024). Organisational design, including communication norms and role clarity, further mediates how leadership identities are enacted within project teams (Walker and Lloyd-Walker, 2024).

The AI-GBL intervention was designed to support structured identity rehearsal in this context. Scenario design emphasised ethically ambiguous and emotionally resonant dilemmas that required learners to

engage in strategic positioning and moral reflection. One illustrative case tasked students with managing a community health project in which a subcontractor offered to accelerate delivery by bypassing safety inspections. Learners were prompted to weigh political pressure against public safety and reflect on how their decisions shaped the kind of leader they were becoming. AI-generated prompts scaffolded this reflection through targeted questions: Who is impacted? What values are prioritised? What kind of leader are you becoming?

These narrative-rich environments functioned less as assessment tools and more as developmental spaces, inviting learners to evaluate, affirm, and reconstruct their emerging leadership identities. This approach is supported by studies highlighting the importance of experiential, ethical, and role-based learning for leadership formation (Prince, 2004; Özsoy and Sezgili, 2024). The VICE framework similarly points to the value of structured reflection and character rehearsal within virtual environments (Goedert et al., 2011).

Empirical data from this study suggest that learners engaged with identity-level reflection. Reflective journals increasingly include values such as trust, accountability, and integrity. Linguistic analysis revealed a shift from individual to collective phrasing (“we decided”, “our team’s values”), echoing principles of identity-based leadership (Haslam et al., 2022). Tutor observations also noted that students adapted decisions mid-scenario in response to feedback, demonstrating growing emotional flexibility and situated judgement.

While identity formation is inherently a long-term, socially embedded process, these early signs of narrative engagement are pedagogically significant. The study did not track development beyond the course, and linguistic indicators remain interpretive. Nonetheless, the findings support the value of treating leadership identity not as an incidental byproduct, but as a deliberate educational aim. Culturally responsive design is also critical. As all scenarios were developed within a Western academic paradigm, further work is needed to explore how identity rehearsal may reflect diverse cultural assumptions around ethics, authority, and leadership.

The learnings from this theme are that thoughtfully designed AI–GBL environments can offer a developmental arena where learners begin to explore who they are when they lead. By anchoring leadership in reflection, ethical reasoning, and narrative construction, such environments move the pedagogical focus from what leaders do to how they interpret, align, and position themselves in uncertain and ethically charged project settings.

3.4. Theme 4 – AI ethics

As artificial intelligence becomes more deeply embedded in project environments, leaders must navigate complex ethical questions concerning transparency, bias, and broader societal impact. However, such issues are still not adequately addressed in project management education, where ethical reasoning is often regarded as peripheral rather than central. In a time shaped by intelligent systems, the ability to evaluate algorithmic advice and consider sustainability trade-offs is essential.

Central to this ability is developing ethical AI literacy. This means the skill to critically analyse AI outputs, recognise underlying assumptions, and evaluate effects for different stakeholders. Ethically aware professionals are not passive recipients of AI suggestions; they examine system design, spot bias, and make decisions that consider context and human values. A virtue ethics framework, based on Aristotelian ideas of character, judgement, and *phronēsis*, provides a solid foundation for fostering such attitudes (Eikeland, 2012).

Ethical leadership in projects rarely involves simple choices. As Dalcher (2024) notes, project dilemmas often reveal tensions between conflicting rights rather than clear-cut right or wrong options. Organisational pressures, including deadlines, KPIs, and stakeholder politics, intensify these tensions. In this setting, ethical competence requires moral courage and the willingness to ask tough questions, challenge

assumptions, and be accountable for long-term outcomes.

The AI–GBL intervention incorporated these dynamics into its scenario design. Instead of guiding learners toward specific answers, it presented ethically complex situations with unpredictable outcomes and conflicting stakeholder interests. Through scaffolded reflection, learners were encouraged to justify their decisions, reflect on their values, and scrutinise AI-provided advice.

Postgraduate journal data show significant engagement. Students identified biases in AI outputs, such as simplified portrayals of stakeholders or overly confident forecasts, and assessed immediate benefits against long-term societal impacts. Notably, 61 % of journals mentioned sustainability, with learners actively comparing algorithmic reasoning to their personal ethical standards. One student observed that a politically convenient decision could jeopardise public safety, illustrating an increasing alignment between ethical considerations and leadership judgment.

These findings reinforce arguments that ethical AI literacy combines technical knowledge with moral judgement, requiring an understanding of how systems operate and discernment about when—and whether—they should be trusted (Floridi, 2021). Incorporating virtue ethics into AI–GBL design emphasised responsibility, character, and reflective action (Eikeland, 2012). In this way, the intervention served not only as a simulation tool but also as a pedagogical space for moral development.

While reflective journals revealed increased ethical sensitivity, they provide limited insight into long-term behavioural change. Future research could incorporate longitudinal follow-up and behavioural observation to evaluate how dispositions transfer into professional settings. Additionally, variation in engagement suggests a need to differentiate levels of ethical reasoning and scaffold accordingly.

Finally, ethical pluralism warrants further exploration. Scenarios were crafted within a Western academic framework, potentially embedding assumptions about authority and individual agency. Although international students participated actively, future iterations should develop more culturally responsive designs to ensure inclusivity and wider ethical relevance.

This theme emphasises a core point: technical skill with AI alone is not enough without ethical judgement. Project leaders must be ready not just to work with AI but also to question it, responsibly interpret its outputs, and stay grounded in human-centred values. AI–GBL environments offer a promising way to achieve this by allowing learners to reflect on what AI can do, what it should do, and who they want to become as they make decisions.

3.5. Theme 5 – motivation

Sustained motivation is widely recognised as vital for deep learning in project management education. However, traditional instruction often finds it difficult to keep learners engaged as they encounter increasing complexity and ambiguity (Csikszentmihalyi and Csikszentmihalyi, 1990). Without psychological momentum, emerging project leaders may lose signals of progression and mastery, weakening their resilience under pressure.

At the heart of this challenge lies the concept of momentum. We define this as the learner’s forward movement marked by intrinsic motivation, sustained engagement, and incremental mastery. This is closely linked to flow theory, which describes a state of deep absorption when task difficulty is optimally matched to skill. Flow fosters persistence, emotional investment, and autonomous learning (Nash, 2025). A recent meta-analysis confirms that flow significantly predicts long-term academic persistence across diverse contexts (Zhang and Wang, 2024).

Self-Determination Theory complements this view, identifying autonomy, competence, and relatedness as drivers of intrinsic motivation (Chapman et al., 2025; Ilies et al., 2017). When these psychological needs are met, learners report greater engagement and well-being. Blended learning studies further show that designs supporting autonomy and feedback sustain motivation over time (Heilporn et al., 2021;

Lin et al., 2019). These insights suggest that momentum is not a secondary benefit but a core mechanism for deep learning and professional growth.

In AI-mediated environments, motivation must be carefully scaffolded. Surface-level gamification, such as points or badges, can undermine autonomy and diminish flow if misaligned with pedagogical goals (Chapman et al., 2025; Huang et al., 2023). Motivation emerges not from novelty, but from purposeful, learner-centred design that aligns with learners' goals, values, and professional aspirations (Bond et al., 2018; Pan et al., 2023).

The AI-GBL model in this study deliberately integrated motivational design. Scenarios provided adaptive challenges, real-time responsiveness, and narrative coherence, conditions that are known to encourage flow. Undergraduate participants often described their experiences as "immersive," "absorbing," and "consistently rewarding." Postgraduates also noted that the AI's dynamic adaptation helped sustain focus and emotional engagement.

These findings indicate that momentum can be fostered through careful alignment of narrative, challenge, and learner agency. While engagement was primarily assessed through self-report, many descriptions reflected flow indicators, such as deep concentration, a loss of time awareness, and intrinsic satisfaction. Such responses align with Nash's (2025) characterisation of flow-driven learners as goal-oriented, feedback-seeking, and resilient.

As discussed in Theme 1, AI-GBL environments can foster meaningful learning when crafted to involve learners at the edge of their competence. Similar design principles apply when maintaining motivational momentum through adaptive challenges. Future research could build on these ideas by using behavioural or physiological indicators, such as gaze-tracking, self-explanation, or argumentation metrics, as proxies for deep engagement (Winje and Løndal, 2020). These tools could triangulate learner experiences and help improve flow-based calibration. Importantly, motivational drivers differ across cultural contexts; therefore, culturally responsive design is essential for ensuring that adaptive systems modify pacing, feedback, and narrative tone to align with diverse learner values and expectations (Beard, 2015).

Theme 5 affirms that motivation must be a core design principle in project leadership education. When calibrated carefully, adaptive AI environments can sustain momentum, foster persistence, and build emotional resilience. These outcomes prepare learners for the more demanding forms of reasoning required to lead amid complexity, setting the stage for the development of adaptive expertise, discussed in the next theme.

3.6. Theme 6 – adaptive expertise

In rapidly changing, AI-integrated project environments, developing adaptive expertise has become a key aim of leadership education. We define this as the ability to apply knowledge flexibly and creatively in unfamiliar or evolving situations, to execute known procedures efficiently, and to innovate, reflect, and problem-solve when facing new challenges.

Traditional teaching methods focusing on best practices or procedural efficiency do not adequately prepare future leaders to manage complexity, volatility, and ethical uncertainty. As automation takes over routine tasks, the human contribution increasingly involves interpretation, judgement, and ethical discernment, skills that are difficult to formalise (Savio and Dewan, 2023; Zarooni and Khatib, 2023).

Contemporary project leadership requires more than just technical expertise. It involves rethinking problems, managing through uncertainty, and acting with moral responsibility under pressure. Huemann and Turner (2024) note that such conditions are not unusual but typical of modern project work. However, many project learning systems still focus on retrospection and error reduction. Lessons-learned routines often reinforce procedural thinking, while enterprise tools promote routine responses that limit adaptive judgement (Marzouk et al., 2022;

McClory et al., 2017).

The concept of productive failure offers a necessary reorientation. Kapur (2013) demonstrates that learners who struggle with complex problems before instruction develop deeper conceptual understanding than those given direct guidance. In this view, failure is not an instructional flaw but a formative condition, epistemically generative and essential to developing expert-level reasoning (Sinha and Kapur, 2021).

The AI-GBL model applied this principle through open-ended, evolving simulations. Learners were not directed towards predefined solutions but had to navigate ambiguity, reversals, and conflicting cues. The AI preserved complexity rather than reducing it, encouraging learners to make iterative decisions and confront dissonance. As one student reflected, "The AI did not let us settle ... later I realised that was the point. I had to lead through ambiguity." In one scenario, students managed a redevelopment project when the AI introduced a late-stage policy shift: part of the site was to be repurposed for emergency refugee housing. Learners faced pressure from local stakeholders opposing the move, while balancing ethical obligations and public accountability. The AI maintained conflicting cues and shifting priorities, forcing students to lead through ambiguity. One noted, "It was not about finding the right answer, it was about learning how to lead when no answer was comfortable."

This trajectory reflects sensemaking theories, in which disorientation comes before insight (Shepherd et al., 2013). Initial frustration led to reflection and cognitive restructuring as learners reassessed assumptions and considered trade-offs. Adaptive expertise, in this sense, is not merely a skillset but a developmental posture and a willingness to reason through change, failure, and ethical uncertainty.

Philosophical work further supports this view. Khanna et al. (2016) argues that failure is not just epistemic but ethical, serving as the space where character, humility, and responsibility develop. In this study, AI-GBL scenarios revealed such ethical tensions, prompting students to consider competing goods, reflect on leadership identity, and resist rushing to conclusions. These moments reflected the moral reasoning and identity work discussed in earlier themes.

At the same time, adaptation was inconsistent. Some learners passively relied on AI outputs or withdrew when ambiguity was emotionally difficult. These patterns reflect concerns about automation-induced disengagement (Savio and Dewan, 2023). As Shepherd et al. (2013) warns, failure must be scaffolded and framed to unlock its developmental potential.

The challenge is to design environments where failure is not simply avoided but made meaningful and impactful. AI-GBL offers an environment where ambiguity is embraced, ethical judgement is practised, and leadership arises not from certainty but through reflective engagement with complexity. In this approach, adaptive expertise shifts from focusing solely on students' knowledge to how they manage and articulate uncertainty as they develop.

4. Discussion

Across the six themes, a coherent and transformative model of project leadership development emerges, shifting focus away from competence-based teaching towards an education rooted in ethical reasoning, adaptive judgement, and reflexive identity. The AI-GBL environment trialled in this study did not merely deliver content; it provided a space where failure, ambiguity, and social complexity drove growth. This change reflects a broader pedagogical imperative: to prepare learners not solely for certainty, but for navigating uncertainty; not merely to manage, but to lead with integrity under conditions that resist control. A unifying insight across themes is the redefinition of failure as a formative experience. Learners faced deliberately ambiguous scenarios within the AI-GBL approach that resisted simple solutions. In Theme 6, for instance, this ambiguity was not an instructional flaw but a purposeful disruption designed to provoke reflection, emotional

dissonance, and iterative sensemaking. Echoing Kapur's (2013) idea of productive failure and Khanna's (2016) ethical framing of failure, these moments acted as developmental thresholds, prompting learners to re-evaluate their decisions and emerging leadership identities.

Narrative identity also became a vital aspect of transformation. As discussed in Theme 3, learners began to craft their own leadership stories, gradually shifting from merely describing their actions to reflecting on who they were becoming. This journey was often driven by morally charged dilemmas where character, rather than procedure, emerged as the primary mode of action. These reflections suggest that leadership development is both cognitive and ontological, involving a process of becoming shaped by emotional and ethical encounters. Themes 2 and 5 further demonstrated that virtues such as humility, courage, and deliberation emerged not as abstract theories but as lived responses to complex problems. Students did not simply learn about these values; they practised them, often under emotional tension. The AI-GBL reinforced this by maintaining ambiguity and providing stakeholder prompts that complicated simple answers. This supports the idea that AI-GBL can act as a co-ordinator of ethical development, not merely a delivery tool.

At a curricular level, these insights support a vital evolution in project management education. As intelligent systems increasingly automate routine tasks, the human domain becomes one of interpretation, ethics, and relational leadership. In this context, AI should not be seen as a threat to education but as a collaborator in developing what cannot be automated: moral reasoning, adaptive thinking, and genuine leadership presence. Moreover, the AI-GBL environment was more than just a simulation; it acted as a philosophical intervention. It demanded that educators assume new roles, not merely as content experts but as designers of complexity and facilitators of ambiguity. This role shift mirrors the broader educational move towards reflexive pedagogy and complexity-aware design.

This research offers both a conceptual and practical guide on how AI-GBL can improve project management education and shape human interactions with emerging intelligent systems. By embedding ethical reasoning, adaptive challenges, and identity development into simulated experiences, AI-GBL promotes the development of future-ready leaders who can navigate ambiguity with confidence and responsibility. This method links technological innovation with humanistic values, emphasising agency, reflection, and moral judgement as vital skills in an AI-enabled world. The six key design principles—relevance, agility, identity, ethics, motivation, and adaptability—provide a cohesive pedagogical framework for educators and practitioners to enhance learning for both professional skills and personal development.

As practitioner-scholars operating at the crossroads of academia, industry, and professional development, our approach was highly reflective. The intervention was shaped by the genuine conditions of classroom practice: time limitations, mixed modalities, and resource constraints. Far from reducing its effectiveness, these constraints grounded the intervention in real-world practicality. We aimed not to perfect the system but to demonstrate a design-informed, ethically grounded, scalable approach to leadership education. The illustrative scenarios in Appendix A (Table 3) provide practical pathways for replication and adaptation.

4.1. Conclusion

This study introduces a new pedagogical paradigm for project leadership education that places ambiguity, ethical reasoning, and identity

construction at the core of the learning experience. By integrating productive failure, stakeholder tension, and adaptive challenge within an AI-enhanced game-based learning environment, it advances beyond procedural training to foster judgement, values, and narrative agency.

Through six interconnected themes, the study shows that leadership is a skill to acquire and a disposition to develop. Learners were not asked to solve problems alone but to interpret them, act ethically, and reflect on their implications for themselves and others. The findings confirm that AI-GBL, based on reflective pedagogy, can support this change by automating leadership and enhancing learners' ability to lead in human and humane ways.

This study guides the design of AI-enhanced game-based learning to improve project management education and enhance key human skills in an AI-driven world. As systems grow more complex and intelligent agents assume larger roles, leadership training must go beyond just procedural knowledge, incorporating uncertainty, reflection, and ethical considerations. AI is now a reality, not a future concept. The key challenge is no longer about whether to use AI, but how to steer its development. AI-GBL offers a vital opportunity to align technology with human values, ensuring that learning systems foster competence and conscience.

4.2. Future design considerations

The AI-GBL intervention effectively promoted ethical, narrative, and adaptive skills but revealed areas for improvement. Some learners experienced emotional fatigue or disengagement when ambiguity remained without support, highlighting the need for more responsive, emotionally aware systems. A promising solution is integrating intelligent pedagogical agents, not to give directives but to act as reflective companions, asking questions, presenting perspectives, or encouraging metacognitive insights. These agents could detect emotions like disengagement or frustration and suggest narrative prompts to reframe learning. Care is needed to maintain ambiguity as a creative force, avoiding its reduction to simple outcomes or automatic resolution.

Future iterations should explore cultural responsiveness in scenario design, ensuring ethical frameworks and leadership models reflect global diversity. Longitudinal research is needed to assess the durability of leadership traits developed through AI-GBL, especially identity commitments and ethical reflexivity beyond the classroom. This study offers a practical model and conceptual foundation for AI-enhanced leadership education, encouraging further dialogue and experimentation across disciplines involving ambiguity, ethics, and human agency.

CRediT authorship contribution statement

Matthew Daniels: Writing – review & editing, Writing – original draft, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Éamonn Kelly:** Writing – original draft, Validation. **Sandra Flynn:** Methodology, Investigation, Conceptualization. **John Kelly:** Writing – review & editing, Validation, Methodology, Investigation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A

Table 3
Illustrative scenarios and thematic alignment in AI-GBL

Scenario Title	Aligned Theme(s)	Brief Description	Pedagogical Purpose
1. The Fast-Track Offer	Ethics, AI Literacy	A subcontractor offers to bypass inspections to meet deadlines. AI presents diverse stakeholder perspectives.	Fosters ethical reasoning, stakeholder negotiation, and critical scrutiny of AI-generated advice.
2. Budget Implosion	Agility	Midway through delivery, the project loses 30 % of funding, requiring urgent re-scoping.	Encourages strategic agility, problem reframing, and adaptive response to volatility.
3. Invisible Stakeholders	Identity, Ethics	A marginalised group excluded from initial plans begins protesting.	Cultivates narrative identity, inclusive leadership, and values-based decision-making.
4. The Algorithm Said So	AI Literacy, Ethics	AI recommends a contractor, but the investigation reveals ethical red flags.	Builds capacity to question algorithmic authority and apply virtue-based reasoning.
5. The Team Collapses	Momentum, Identity	Team conflict derails momentum: AI offers reflective prompts mid-scenario.	Reinforces emotional regulation, metacognitive skills, and persistence under stress.
6. No Right Answer	Adaptability, Productive Failure	Learners face an ambiguous decision with no clear solution.	Stimulates leadership judgement, tolerance for uncertainty, and learning through failure.

Data availability

Data will be made available on request.

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